

ARUNACHALAM BARATHIDASAN

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Education

Illinois Institute of Technology

Master of Computer Science

August 2023 – May 2025

Chicago, Illinois

Karunya Institute of Technology and Sciences

Bachelor of Technology in Computer Science and Engineering

July 2016 – May 2020

India

Skills

Programming & Scripting: Python, JavaScript, HTML/CSS, SQL, Unix/Bash.

AI, ML & Autonomous Agents: Reinforcement Learning (RL), LLM Orchestration (GPT-4o, DeepSeek, MiMo-7B-RL), LLM Deployment & Serving, VLM/LLM Verification Pipelines, Scikit-learn, ONNX.

Web & Software Development: Django, FastAPI, React, React Native, Node.js, PyQt6, GraphQL, Playwright, PyAutoGUI.

Databases & Data Engineering: PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, Apache Kafka, Supabase, Schema Design, ETL Pipelines, Workflow Orchestration.

Cloud & DevOps: AWS (EC2, S3, SQS, EMR, SageMaker, Lightsail), GCP (Cloud SQL), Docker, Kubernetes, Git/GitHub.

Security & Blockchain: OAuth, JWT, RBAC, PGP Encryption, Polygon Network, Cryptocurrency APIs, Consensus Algorithms.

Architecture & Systems: Distributed Observability, Microservices, High-Frequency Trading (HFT) concepts, Data Visualization (DAG/Waterfall).

Design & Management: Figma, Adobe XD, Agile Methodologies, Jira, Notion, Adobe Creative Suite.

Experience

COSMOS Research Center

April 2026 – Present

Data Engineer, Research Engineering Team

Little Rock, AR

- Build and maintain **Node Flow**, an internal workflow orchestration platform for research data pipelines, covering data acquisition, preprocessing, graph construction, metadata enrichment, statistical analysis, and visualization generation.
- Design ingestion and preprocessing pipelines over large-scale datasets in **MongoDB** to support downstream research systems.
- Support deployment and applied use of **LLMs** across internal tooling, including document parsing and evidence-grounded research synthesis systems.
- Develop and operate pipelines on Linux research servers via SSH, working across the full lifecycle from data acquisition to synthesis and visualization.

LearnChain

Aug 2025 – Dec 2025

Co-founder & Lead Software Engineer

Chicago, USA

- Architected and built the **LearnChain Platform** and **Cosmic Desktop App** using **Python** and **PyQt6**, delivering scalable AI-assisted workflows for enterprise onboarding.
- Designed high-throughput **ETL pipelines** using **AWS S3 and SQS** to process unstructured documents asynchronously, significantly reducing ingestion latency.
- Engineered three full-stack web applications (LearnChain, Cosmic, and Open Agent Tracer) utilizing **Python (Django)** for robust backend architectures and seamless user-facing flows.
- Implemented complex **Role-Based Access Control (RBAC)** and secure authentication flows (OAuth) across web and desktop clients.
- Optimized database performance by designing normalized schemas in **PostgreSQL** and implementing targeted indexing strategies for query efficiency.
- Containerized microservices using **Docker** and automated deployments to **AWS Lightsail**, ensuring consistent environments across development and production.

Triz AI

Sept 2021 – Jun 2023

Software Engineer

Chennai, India

- Led the development of a cross-platform mobile application using **React Native** and **Node.js**, achieving 98% crash-free sessions in production.
- Integrated AI-driven task allocation algorithms that improved user workflow efficiency by 40%.
- Designed and implemented the core application architecture and database schema in **PostgreSQL**.

Valeo India Pvt Ltd

Jan 2020 – Dec 2020

Infrastructure Security Intern

Chennai, India

- Implemented secure end-to-end PGP encryption for email communication across 1000+ employees.
- Streamlined remote access authentication protocols, improving system response times during high-traffic periods.

Projects

- Coppr – Copper Parser** | *Python, LLM/VLM Pipelines, OCR, Supabase* | copprlab.com **2026**
- Built the fail-closed datasheet parser for Coppr, an agentic **EDA** system that converts a natural-language prompt into a verified component, **KiCad** design, and manufacturable Gerbers.
 - Designed an 8-stage extraction pipeline (78 Python modules) that arbitrates between three independent witnesses – a geometric extractor, a VLM/OCR pixel witness, and an independent table-structure model – to bind every specification value to a traceable source location.
 - Implemented a deterministic evaluation and gating layer with bounded remediation loops, plus an **LLM**-based pre-publication reviewer, to block unverifiable data before it reaches production.
 - Published verified records to **Supabase** over PostgREST with idempotent, content-fingerprinted writes and immutable part revisioning.
- Open Agent Tracer** | *Python, React, Reinforcement Learning, AWS* | oat.thelearnchain.com **Feb 2026**
- Engineered a low-latency observability SDK utilizing **monkey patching** to automatically instrument **multimodal AI agents**, capturing high-fidelity telemetry including cost, latency, and LLM token streams.
 - Designed a centralized control plane featuring a dedicated flow page to map entire agent workflows, utilizing **DAG** and **Waterfall** visualizations to trace non-deterministic execution paths and tool invocation graphs.
 - Built an **LLM model comparison arena** to benchmark performance, and implemented a self-correcting control loop using **Reinforcement Learning (RL)** to analyze execution failures and dynamically adjust retry strategies.
- Vision-Based Browser Agent** | *Python, Playwright, MiMo-7B-RL, Computer Vision* **Jan 2026**
- Built a high-performance, vision-first autonomous agent capable of executing complex web tasks (e.g., multi-leg flight search) in under 60 seconds.
 - Engineered a hybrid navigation system (90% Vision / 10% DOM) utilizing **MiMo-7B-RL** for GUI grounding and **Playwright** for action execution.
 - Developed a novel automation architecture to handle obfuscated interfaces, achieving high success rates on dynamic Single Page Applications (SPAs).
- Polymarket Algorithmic Trading Bot** | *Python, AWS EC2, Polygon, REST APIs* **Jan 2026**
- Engineered a Python-based algorithmic trading bot for Polymarket weather arbitrage, utilizing the Gamma API for market discovery and py-clob-client for automated order execution on the Polygon network.
 - Integrated real-time data pipelines from NOAA and Open-Meteo APIs, targeting 12–36 hour forecast windows to identify and execute on a minimum 10% statistical edge.
 - Implemented rigorous automated risk management protocols deployed 24/7 on AWS EC2, including a 15% daily drawdown kill switch and strict position sizing limits.
 - Designed a low-latency "Commander Architecture" concept incorporating Rust for sub-10ms execution loops in high-frequency crypto trading (HFT) strategies.
- Cosmic** | *Django, Supabase, AWS Lightsail, AWS S3, redis* | cosmic.thelearnchain.com **August 2025**
- Deployed a dynamic web platform on AWS Lightsail, utilizing AWS S3 for high-availability static and media asset management
 - Secured user data persistence by integrating Supabase and established robust authentication workflows using Django AllAuth
 - Refactored the session architecture from Redis caching to Django's native handling, significantly reducing infrastructure overhead and complexity.
 - Project management and version control facilitated with Notion and Git.
- Learnchain** | *Django, Supabase, AWS Lightsail, HTML, tailwind CSS, Javascript* | thelearnchain.com **April 2025**
- Engineered a dynamic Django site with robust backend validation.
 - Orchestrated the seamless integration of Google's Cloud SQL capturing user data into PostgreSQL database.
 - Authenticated users using Firebase.
 - Used AWS Lightsail to host and deploy the website for scalable and cost-effective cloud hosting.
 - Project management and version control facilitated with Notion and Git.
- Data Preparation and Analysis** | *Python, ONNX, Scikit-learn, Matplotlib, Machine Learning* **Nov 2024**
- **Data Analysis:** Engineered machine learning pipeline for 1.2 million record dataset in python.
 - **Data Processing:** Preprocessed data by applying scaling, normalization and feature extraction (PCA, recursive feature elimination).
 - **Machine Learning:** Trained Naive Bayes, Decision Tree and KNN reaching model accuracy of 72%.
 - **Data Visualization:** Visualized using various plots in Matplotlib Library.
 - **Model Deployment:** Converted Machine Learning models to ONNX format for cross-platform integration.
- Datasys Coin** | *Python, Hashlib, Sockets, BeautifulSoup4, Blockchain* **Nov 2023**
- **Architecture:** Deployed basic architecture involving several individual components such as validator, metronome, monitor, pool, blockchain, wallet etc. Client-Server architecture using Socket programming in python.
 - **Blockchain Development:** Built a cryptocurrency with consensus algorithms such as proof of work, proof of space.
 - **Security and Encryption:** Generated 10,000+ SHA-256 hashes ensuring blockchain security.
 - **Network Infrastructure:** Peer to Peer transaction validation system.